

SUPPLEMENTARY MATERIAL

Adaptive Nonlinear Vector Autoregression: Robust Forecasting for Noisy Chaotic Time Series

Sherkhon Azimov, Susana López-Moreno, Eric Dolores-Cuenca, Sieun Lee, Jae-Il Kwon
and Sangil Kim

Supplementary Example

We illustrate our adaptive NVAR model using a simple 3-dimensional time series with delay embedding and joint feature learning.

- (i) Dimension of the state vector: $d = 3$.
- (ii) Number of delays: $k = 2$.
- (iii) Neural network output dimension: $d_{\mathcal{NN},i} = dk(dk + 1)/2$, to match the number of fixed nonlinear features in NVAR. Alternatively, $d_{\mathcal{NN},i} = cdk$, with c being a tunable constant.
- (iv) Input dimension: $dk + d_{\mathcal{NN},i} = 6 + 21 = 27$.
- (v) Number of training points: $n = 5$ ($I_{\text{train}} = \{2, 3, 4, 5, 6\}$ are valid training indices for $k = 2$).
- (vi) For each $i \in I_{\text{train}}$, we use $H_{\text{lin},i} = X_i \oplus X_{i-1} \in \mathbb{R}^6$.

Let $X_i \in \mathbb{R}^d$ for $i = \{1, \dots, 10\}$ be our toy series dataset, where $d = 3$. The following table lists the components of X_i .

i	1	2	3	4	5	6	7	8	9	10
$x_{1,i}$	0.1	0.2	0.3	0.5	0.4	0.7	0.6	0.8	0.9	0.5
$x_{2,i}$	0.3	0.1	0.6	0.2	0.5	0.1	0.4	0.6	0.5	0.2
$x_{3,i}$	0.2	0.4	0.1	0.3	0.6	0.2	0.3	0.2	0.9	0.2

Table 1: Values of $x_{j,i}$ for $j = 1, 2, 3$ and $i = 1, \dots, 10$.

We now construct the linear feature vectors and their corresponding targets for training:

i	$H_{\text{lin},i}$	Target X_{i+1}
2	$X_2 \oplus X_1$	X_3
3	$X_3 \oplus X_2$	X_4
4	$X_4 \oplus X_3$	X_5
5	$X_5 \oplus X_4$	X_6
6	$X_6 \oplus X_5$	X_7

Table 2: Linear combination inputs $H_{\text{lin},i}$ and their target outputs X_{i+1} .

The feature vectors are then in the following vector spaces:

1. The linear feature vector $H_{\text{lin},i} \in \mathbb{R}^{dk} = \mathbb{R}^6$.
2. The neural network (nonlinear) feature vector $H_{\mathcal{NN},i} = \mathcal{NN}(H_{\text{lin},i}; \theta) \in \mathbb{R}^{d_{\mathcal{NN},i}} = \mathbb{R}^{21}$.
3. The total feature vector, defined by concatenating the linear and nonlinear features $H_{\text{total},i} = H_{\text{lin},i} \oplus H_{\mathcal{NN},i} \in \mathbb{R}^{dk+d_{\mathcal{NN},i}} = \mathbb{R}^{27}$.

The neural network $\mathcal{NN}(H_{\text{lin},i}; \theta)$ is defined as a simple MLP with one hidden layer and a tanh activation function. Let the hidden layer have $h = 24$ units. The MLP transformation is given by:

$$H_{\mathcal{NN},i} = \mathcal{NN}(H_{\text{lin},i}; \theta) = W \cdot \tanh(W_{\text{in}} H_{\text{lin},i}),$$

where $W_{\text{in}} \in \mathbb{R}^{h \times (dk)} = \mathbb{R}^{24 \times 6}$ and $W \in \mathbb{R}^{d_{\mathcal{NN},i} \times h} = \mathbb{R}^{21 \times 24}$.

After concatenating $H_{\text{lin},i}$ and $H_{\mathcal{NN},i}$ into $H_{\text{total},i}$, we perform the readout layer as follows:

$$\hat{Y}_i = \hat{X}_{i+1} - \hat{X}_i = W_{\text{out}} H_{\text{total},i}.$$

Thus, the parameter set θ is $\theta = \{W_{\text{in}}, W, W_{\text{out}}\}$.

The mean squared error loss

$$L(\theta) = \frac{1}{5} \sum_{i=2}^6 \|W_{\text{out}} H_{\text{total},i} - Y_i\|_2^2 = \frac{1}{5} \sum_{i=2}^6 \|W_{\text{out}} [H_{\text{lin},i} \oplus \mathcal{NN}(H_{\text{lin},i}; \theta)] - Y_i\|_2^2$$

is used to compute the gradients with respect to both θ .

After completing training, we evaluate the model using the remaining points, referred to as test points. For each prediction, we use the trained W_{out} and $H_{\text{total},i}$.

$$\hat{Y}_7 = W_{\text{out}} H_{\text{total},7}, \text{ where } H_{\text{total},7} = H_{\text{lin},7} \oplus H_{\mathcal{NN},7}$$

$$\hat{Y}_8 = W_{\text{out}} H_{\text{total},8}, \text{ where } H_{\text{total},8} = H_{\text{lin},8} \oplus H_{\mathcal{NN},8}$$

and so on.

Supplementary Tables

Model	Hyperparameter	Value
Standard NVAR	Skipping parameter (s)	1
Adaptive NVAR	Skipping parameter (s)	1
Adaptive NVAR (Lorenz-96)	Mini-batch size	1024
ESN	Washout length (test/validation)	500 (MG), 100 (L63)
HESN	Washout length (test/validation)	500 (MG), 100 (L63)

Table 3: Fixed hyperparameters used in each model.

Noise Level	Dataset	Horizon 25	Horizon 50	Horizon 75	Horizon 100
Adaptive NVAR					
Noise free	Shared (All)	0.0057 ± 0.0015	0.0079 ± 0.0025	0.0103 ± 0.0034	0.0133 ± 0.0047
Low noise	Dataset 1	0.0114 ± 0.0004	0.0138 ± 0.0006	0.0163 ± 0.0008	0.0195 ± 0.0012
	Dataset 2	0.0135 ± 0.0006	0.0170 ± 0.0010	0.0206 ± 0.0138	0.0249 ± 0.0017
	Dataset 3	0.0122 ± 0.0004	0.0152 ± 0.0007	0.0187 ± 0.0011	0.0223 ± 0.0013
Moderate noise	Dataset 1	0.0225 ± 0.0006	0.0271 ± 0.0007	0.0326 ± 0.0010	0.0389 ± 0.0012
	Dataset 2	0.0214 ± 0.0004	0.0252 ± 0.0005	0.0306 ± 0.0006	0.0387 ± 0.0011
	Dataset 3	0.0220 ± 0.0006	0.0261 ± 0.0007	0.0304 ± 0.0009	0.0349 ± 0.0014
High noise	Dataset 1	0.0304 ± 0.0005	0.0375 ± 0.0005	0.0455 ± 0.0007	0.0540 ± 0.0010
	Dataset 2	0.0321 ± 0.0004	0.0390 ± 0.0004	0.0456 ± 0.0006	0.0530 ± 0.0009
	Dataset 3	0.0329 ± 0.0003	0.0404 ± 0.0007	0.0479 ± 0.0008	0.0550 ± 0.0008
NG-RC					
Noise free	Shared (All)	0.0385	0.0529	0.0636	0.0729
Low noise	Dataset 1	0.0240	0.0318	0.0390	0.0471
	Dataset 2	0.0253	0.0320	0.0389	0.0476
	Dataset 3	0.0256	0.0331	0.0397	0.0479
Moderate noise	Dataset 1	0.0340	0.0427	0.0537	0.0669
	Dataset 2	0.0329	0.0404	0.0504	0.0637
	Dataset 3	0.0345	0.0434	0.0548	0.0679
High noise	Dataset 1	0.0459	0.0573	0.0697	0.0833
	Dataset 2	0.0510	0.0579	0.0668	0.0770
	Dataset 3	0.0511	0.0641	0.0762	0.0884
ESN					
Noise free	Shared (All)	0.0321 ± 0.0006	0.0319 ± 0.0010	0.0318 ± 0.0015	0.0318 ± 0.0018
Low noise	Dataset 1	0.0404 ± 0.0223	0.0432 ± 0.0283	0.0459 ± 0.0319	0.0497 ± 0.0350
	Dataset 2	0.0374 ± 0.0100	0.0420 ± 0.0204	0.0472 ± 0.0302	0.0535 ± 0.0377
	Dataset 3	0.0532 ± 0.0321	0.0611 ± 0.0391	0.0759 ± 0.0541	0.0861 ± 0.0634
Moderate noise	Dataset 1	0.0330 ± 0.0024	0.0348 ± 0.0041	0.0371 ± 0.0064	0.0418 ± 0.0097
	Dataset 2	0.0365 ± 0.0024	0.0365 ± 0.0036	0.0375 ± 0.0052	0.0411 ± 0.0078
	Dataset 3	0.0587 ± 0.0105	0.0784 ± 0.0179	0.0928 ± 0.0202	0.1035 ± 0.0188
High noise	Dataset 1	0.0356 ± 0.0028	0.0385 ± 0.0042	0.0435 ± 0.0055	0.0500 ± 0.0074
	Dataset 2	0.0367 ± 0.0059	0.0418 ± 0.0087	0.0479 ± 0.0114	0.0567 ± 0.0144
	Dataset 3	0.0540 ± 0.0191	0.0676 ± 0.0357	0.0768 ± 0.0444	0.0847 ± 0.0484
HESN					
Noise free	Shared (All)	0.0390 ± 0.0093	0.0470 ± 0.0217	0.0559 ± 0.0397	0.0686 ± 0.0684
Low noise	Dataset 1	0.0336 ± 0.0022	0.0349 ± 0.0035	0.0362 ± 0.0047	0.0386 ± 0.0061
	Dataset 2	0.0365 ± 0.0144	0.0394 ± 0.0181	0.0420 ± 0.0213	0.0451 ± 0.0237
	Dataset 3	0.0338 ± 0.0040	0.0353 ± 0.0053	0.0367 ± 0.0059	0.0387 ± 0.0066
Moderate noise	Dataset 1	0.0383 ± 0.0077	0.0435 ± 0.0109	0.0497 ± 0.0150	0.0567 ± 0.0184
	Dataset 2	0.0340 ± 0.0010	0.0362 ± 0.0021	0.0386 ± 0.0032	0.0421 ± 0.0045
	Dataset 3	0.0341 ± 0.0071	0.0371 ± 0.0100	0.0407 ± 0.0127	0.0449 ± 0.0137
High noise	Dataset 1	0.0395 ± 0.0063	0.0446 ± 0.0077	0.0510 ± 0.0102	0.0581 ± 0.0124
	Dataset 2	0.0364 ± 0.0055	0.0385 ± 0.0077	0.0414 ± 0.0106	0.0452 ± 0.0126
	Dataset 3	0.0445 ± 0.0159	0.0501 ± 0.0214	0.0556 ± 0.0252	0.0610 ± 0.0278

Table 4: Robustness analysis of Adaptive NVAR, NG-RC, ESN, and HESN models evaluated on three distinct Mackey-Glass datasets. Horizons 25–100 are reported across four noise regimes (Noise-free, Low, Moderate, and High). For stochastic models, results indicate mean $\pm$ standard deviation. NG-RC performance is reported as a scalar value due to its deterministic execution. (Rounded)

Noise Level	Dataset	Horizon 25	Horizon 50	Horizon 75	Horizon 100
Adaptive NVAR					
Noise free	Shared (All)	0.0064 ± 0.0009	0.0166 ± 0.0039	0.0303 ± 0.0088	0.0610 ± 0.0306
Low noise	Dataset 1	0.0915 ± 0.0117	0.1132 ± 0.0106	0.1297 ± 0.0120	0.1613 ± 0.0195
	Dataset 2	0.0787 ± 0.0034	0.1009 ± 0.0037	0.1292 ± 0.0082	0.1703 ± 0.0126
	Dataset 3	0.0889 ± 0.0028	0.1148 ± 0.0053	0.1457 ± 0.0070	0.2410 ± 0.0148
Moderate noise	Dataset 1	0.1541 ± 0.0033	0.1906 ± 0.0031	0.2874 ± 0.0124	0.3730 ± 0.0118
	Dataset 2	0.1435 ± 0.0038	0.1799 ± 0.0094	0.2457 ± 0.0175	0.3775 ± 0.0244
	Dataset 3	0.2331 ± 0.0117	0.2777 ± 0.0139	0.3651 ± 0.0327	0.4624 ± 0.0428
High noise	Dataset 1	0.3305 ± 0.0156	0.3636 ± 0.0194	0.4408 ± 0.0241	0.5160 ± 0.0297
	Dataset 2	0.2330 ± 0.0128	0.3196 ± 0.0178	0.3992 ± 0.0392	0.4768 ± 0.0413
	Dataset 3	0.3216 ± 0.0062	0.3747 ± 0.0210	0.4883 ± 0.0275	0.6110 ± 0.0367
NG-RC					
Noise free	Shared (All)	0.0023	0.0054	0.0081	0.0102
Low noise	Dataset 1	0.1140	0.2474	0.4119	0.5295
	Dataset 2	0.1589	0.2501	0.3794	0.5199
	Dataset 3	0.1017	0.1795	0.2646	0.3219
Moderate noise	Dataset 1	0.2267	0.3103	0.4653	0.6536
	Dataset 2	0.1631	0.3108	0.4590	0.5986
	Dataset 3	0.2428	0.3691	0.5460	0.6981
High noise	Dataset 1	0.2546	0.4216	0.6366	0.8062
	Dataset 2	0.3041	0.4861	0.6913	0.8619
	Dataset 3	0.2956	0.5347	0.7518	0.9183
ESN					
Noise free	Shared (All)	0.1514 ± 0.0040	0.1944 ± 0.0379	0.2852 ± 0.0934	0.4207 ± 0.1547
Low noise	Dataset 1	0.1525 ± 0.0072	0.1724 ± 0.0170	0.2077 ± 0.0302	0.2507 ± 0.0432
	Dataset 2	0.1658 ± 0.0118	0.2106 ± 0.0207	0.2572 ± 0.0297	0.3186 ± 0.0422
	Dataset 3	0.1773 ± 0.0134	0.2125 ± 0.0204	0.2525 ± 0.0353	0.3500 ± 0.0390
Moderate noise	Dataset 1	0.1987 ± 0.0147	0.2893 ± 0.0381	0.3931 ± 0.0610	0.4955 ± 0.0789
	Dataset 2	0.1853 ± 0.0118	0.2403 ± 0.0324	0.3766 ± 0.0603	0.5044 ± 0.0925
	Dataset 3	0.2127 ± 0.0212	0.3151 ± 0.0347	0.4200 ± 0.0630	0.5281 ± 0.0854
High noise	Dataset 1	0.2808 ± 0.0251	0.4061 ± 0.0504	0.5894 ± 0.0748	0.7412 ± 0.0831
	Dataset 2	0.2123 ± 0.0255	0.3419 ± 0.0507	0.4400 ± 0.0666	0.5280 ± 0.0691
	Dataset 3	0.2466 ± 0.0172	0.3296 ± 0.0286	0.4625 ± 0.0407	0.6065 ± 0.0565
HESN					
Noise free	Shared (All)	0.1965 ± 0.0106	0.3008 ± 0.0406	0.5066 ± 0.0570	0.6384 ± 0.0795
Low noise	Dataset 1	0.1721 ± 0.0043	0.2456 ± 0.0175	0.3904 ± 0.0340	0.4967 ± 0.0563
	Dataset 2	0.1772 ± 0.0140	0.2236 ± 0.0186	0.2846 ± 0.0354	0.3541 ± 0.0534
	Dataset 3	0.1784 ± 0.0149	0.2123 ± 0.0258	0.2668 ± 0.0358	0.3549 ± 0.0490
Moderate noise	Dataset 1	0.1809 ± 0.0108	0.2711 ± 0.0363	0.3772 ± 0.0724	0.5122 ± 0.0952
	Dataset 2	0.2061 ± 0.0173	0.3647 ± 0.0594	0.5548 ± 0.0738	0.7211 ± 0.0859
	Dataset 3	0.2334 ± 0.0197	0.3615 ± 0.0588	0.5230 ± 0.1041	0.6704 ± 0.1206
High noise	Dataset 1	0.2828 ± 0.0360	0.4290 ± 0.0740	0.6075 ± 0.1096	0.7367 ± 1.1151
	Dataset 2	0.2485 ± 0.0282	0.4270 ± 0.0784	0.5763 ± 0.1044	0.6854 ± 0.1208
	Dataset 3	0.3422 ± 0.0185	0.5651 ± 0.0477	0.7778 ± 0.0702	0.8882 ± 0.0726

Table 5: Robustness analysis of Adaptive NVAR, NG-RC, ESN, and HESN models evaluated on three distinct Lorenz 63 datasets. Horizons 25–100 are reported across four noise regimes (Noise-free, Low, Moderate, and High). For stochastic models, results indicate mean $\pm$ standard deviation. NG-RC performance is reported as a scalar value due to its deterministic execution. (Rounded)

Dataset	Noise Level	25 Steps	50 Steps	75 Steps	100 Steps
Dataset 1	Noise free	0.0306 ± 0.0040	0.0612 ± 0.0091	0.1029 ± 0.0158	0.1597 ± 0.0236
	Low noise	0.0859 ± 0.0011	0.1475 ± 0.0023	0.2286 ± 0.0036	0.3254 ± 0.0048
	Moderate noise	0.1684 ± 0.0056	0.2719 ± 0.0120	0.3873 ± 0.0162	0.5009 ± 0.0173
	High noise	0.2367 ± 0.0015	0.3664 ± 0.0034	0.5004 ± 0.0047	0.6172 ± 0.0053
Dataset 2	Noise free	0.0306 ± 0.0040	0.0612 ± 0.0091	0.1029 ± 0.0158	0.1597 ± 0.0236
	Low noise	0.0854 ± 0.0009	0.1464 ± 0.0019	0.2270 ± 0.0030	0.3232 ± 0.0040
	Moderate noise	0.1682 ± 0.0050	0.2712 ± 0.0109	0.3862 ± 0.0149	0.4998 ± 0.0161
	High noise	0.2357 ± 0.0010	0.3642 ± 0.0023	0.4976 ± 0.0036	0.6145 ± 0.0045
Dataset 3	Noise free	0.0306 ± 0.0040	0.0612 ± 0.0091	0.1029 ± 0.0158	0.1597 ± 0.0236
	Low noise	0.0858 ± 0.0013	0.1474 ± 0.0027	0.2287 ± 0.0041	0.3254 ± 0.0053
	Moderate noise	0.1693 ± 0.0063	0.2733 ± 0.0136	0.3892 ± 0.0183	0.5029 ± 0.0196
	High noise	0.2366 ± 0.0014	0.3664 ± 0.0031	0.5005 ± 0.0046	0.6180 ± 0.0058

Table 6: Forecasting RMSE (mean $\pm$ standard deviation) of the Adaptive NVAR model under different noise levels and datasets on Lorenz 96 ($N = 100$).

Model	System	Feature Formula	Parameter Values	Feature Size (m)
Standard NVAR	Mackey–Glass	$dk(dk + 1)/2$	$d = 1, k = 10$	55
Standard NVAR	Lorenz–63	$dk(dk + 1)/2$	$d = 3, k = 10$	4,515
Standard NVAR	Lorenz–96	$dk(dk + 1)/2$	$d = 100, k = 20$	2,001,000
Adaptive NVAR	Mackey–Glass	$dk(dk + 1)/2$	$d = 1, k = 10$	55
Adaptive NVAR	Lorenz–63	$dk(dk + 1)/2$	$d = 3, k = 10$	4,515
Adaptive NVAR	Lorenz–96	$c dk$	$n = 10, d = 100, k = 20$	20,000

Table 7: Examples: Nonlinear feature dimensionality for the NVAR and Adaptive NVAR models. Here, d is the system dimension, k the number of delay taps, and n the number of hidden neurons per delayed variable in the Adaptive NVAR. For high-dimensional systems such as Lorenz–96, the Adaptive NVAR replaces the polynomial feature expansion with a shallow MLP of output size $m = n dk$.